illustrate in Figure 8.5. Using this second dimension, a doctor can operate on your body by reaching directly inside your exposed interior. Weird! In time, Linebeings, no doubt, would develop a skinlike cover to shield the newly exposed interior of their bodies from contact with the outside world. And moreover, they would undoubtedly evolve into beings with length as well as breadth: Flatbeings sliding along the two-dimensional Garden-hose universe as illustrated in Figure 8.6. If the circular dimension were to grow very large indeed, this two-dimensional universe would be closely akin to Abbott's Flatland—an imaginary two-dimensional world Abbott suffused with a rich cultural heritage and even a satirical caste system based upon one's geometrical shape. Whereas it's hard to imagine anything interesting happening in Lineland—there is just not enough room—life on a Garden-hose becomes replete with possibilities. The evolution from one to two observably large space dimensions is dramatic.

And now the refrain: Why stop there? The two-dimensional universe might itself have a curled-up dimension and therefore secretly be three-dimensional. We can illustrate this with Figure 8.4, so long as we recognize that we are now imagining that there are only two extended space dimensions (whereas when we first introduced this figure we were imagining the flat grid to represent three extended dimensions). If the circular dimension should expand, a two-dimensional being would find itself in a vastly new world in which movement is not limited just to left-right and back-forth along the extended dimensions. Now, a being can also move in a third dimension —the "up-down" direction along the circle. In fact, if the circular dimension were to grow to a large enough size, this could be our three-dimensional universe. We do not know at present whether any of our three spatial dimensions extends outward forever, or in fact curls back on itself in the shape of a giant circle, beyond the range of our most powerful telescopes. If the circular dimension in Figure 8.4 got big enough—billions of light-years in extent—the figure could very well be a drawing of our world.

But the refrain replays: Why stop there? This takes us to Kaluza's and Klein's vision: that our three-dimensional universe might have a previously unanticipated curled-up fourth spatial dimension. If this striking possibility, or its generalization to numerous curled-up